

PROJECT REPORT ON
“IDENTIFYING HIGH-RISK EPIDEMIC ZONES IN
INDIA: A DATA-DRIVEN RISK SCORE AND
CLUSTERING APPROACH”

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE**CERTIFICATE**

This is to certify that the project entitled “IDENTIFYING HIGH-RISK EPIDEMIC ZONES IN INDIA: A DATA-DRIVEN RISK SCORE AND CLUSTERING APPROACH” is a part of the Integrated Professional Core Course Fundamentals of Machine Learning. This report has been submitted by Liesel Doreen Dsouza (NNM24AD035), Shreya P J (NNM24AD060), Shreya S Salian (NNM24AD061) and Sunidhi S S (NNM24AD067) of II year B. Tech, bonafide students of NMAM Institute of Technology, Nitte, the project work is carried out during the academic year 2025-2026.

Project Guide**Course Coordinator**

Table of Contents

Chapter	Page No.
Front page	1
Certificate Page	2
Abstract	4
1. Introduction	5
2. Literature Survey	6
3. Problem Statement	18
4. Objectives	19
4.1 Data acquisition and preprocessing	19
4.2 Risk score computation	19
4.3 Clustering-based zone identification	19
4.4 Supervised classification for risk prediction	19
4.5 Visualization and reporting	20
5. Dataset Description	21
5.1 Overview of datasets used	21
5.2 EpiClim dataset	22
5.3 IMD rainfall dataset	24
6. Methodology	25
6.1 Data preprocessing	26
6.2 Risk score computation	30
6.3 Clustering-based risk zone identification	33
6.4 Supervised classification for risk prediction	37
6.5 Justification for chosen techniques	38
7. Implementation	40
9. Result	47
10. Conclusion	54
11. References	55

Abstract

India has experienced a steady increase in the frequency and severity of epidemic outbreaks in recent years. This trend is driven by multiple interconnected factors, such as climate change, rapid urbanization, high population density, and increased human mobility. Variations in climate conditions, such as temperature and rainfall, significantly influence the spread of infectious and vector-borne diseases such as Malaria, dengue etc.

These challenges highlight the importance of identifying epidemic-prone regions at an early stage to reduce potential public health impacts. However, traditional disease surveillance systems mainly act after a problem has already started, rather than preventing it beforehand. To address this limitation, this project proposes a proactive, data-driven framework for identifying high-risk epidemic zones across districts in India. The project uses data and analytical methods to predict risk in advance, rather than waiting for outbreaks to happen.

The framework integrates a diverse set of indicators related to climatic conditions, demographic characteristics, healthcare infrastructure, and historical disease incidence. These indicators are merged into one single numerical score that represents how risky an area is in terms of epidemic outbreaks. In addition to risk scoring, unsupervised clustering techniques are applied to group districts with similar risk profiles. Machine learning methods are used to automatically group districts based on similarities in their risk-related data. Unsupervised means the model is not told in advance which districts are high or low risk, it finds patterns on its own. This clustering-based approach enables the classification of regions into low, medium, and high-risk zones, providing a clearer spatial understanding of epidemic risk across the country. Such zoning can support targeted surveillance and timely public health interventions. Overall, this study demonstrates the effectiveness of data-driven methodologies in supporting evidence-based public health decision-making and strengthening India's capacity to anticipate and respond to epidemic threats.

Chapter 1

Introduction

India remains highly vulnerable to epidemic outbreaks due to its large population, various climatic conditions, rapid urbanization etc. In recent times, the frequency and intensity of infectious disease outbreaks such as dengue, malaria, and COVID-19 have increased significantly. Climate change has further intensified this risk by altering temperature patterns, rainfall variability, humidity levels, and extreme weather events. These factors make epidemic preparedness and response an increasingly complex public health challenge in India.

Traditional epidemic surveillance systems in India largely rely on historical case reporting and reactive response mechanisms. While such approaches are useful for outbreak detection, they often fail to provide early warnings or spatial prioritization needed for proactive intervention. The absence of a proper system to assess disease risk at the district level, health authorities face many difficulties. They are unable to decide where to send medical staff, medicines, and other resources on time. It also becomes hard to focus surveillance and monitoring in areas that are more likely to face an outbreak.

Therefore, there is a growing need for analytical frameworks that can identify high-risk epidemic zones before outbreaks escalate. Identifying areas that are at high risk of disease outbreaks at an early stage is very important to reduce the number of cases and deaths. When high risk regions are known in advance, the government can take preventive steps. These kinds of early actions help prevent outbreaks from becoming severe. Such a proactive approach is especially necessary in India because healthcare facilities and access to medical services are not the same in all regions. The work aims to: identify high-risk epidemic zones in India using a data-driven risk score and clustering approach.

Chapter 2

Literature Surveys

2.1 EpiClim: Weekly District-Wise all-India multi-epidemics Climate-Health Dataset for accelerated GeoHealth research

The EpiClim dataset covers district level outbreak records across India from 2009 to 2022. The idea is to track how diseases move across geography and time, and pair that with what the climate was doing at the same moment. Every record tells us when and where an outbreak happened, how many people were affected, how many died, and what the temperature, rainfall, and vegetation cover looked like in that district at that time.

Most of the epidemiological data comes from India's Integrated Disease Surveillance Programme(IDSP), which puts out weekly outbreak reports covering diseases like Dengue, Malaria, Cholera, and Acute Diarrhoeal Disease. To make the data usable for spatial analysis, each district was matched to its latitude and longitude coordinates, so patterns can be mapped rather than just listed which helped us map out risk zones in our clustering models.

The finding in the data is that climate and disease are closely linked. When it gets hotter and wetter, case counts tend to follow. This holds across diseases and regions, which makes temperature and precipitation some of the most useful predictors to work with [1].

2.2 Complex heatmap visualization

Heatmaps provide a visual representation of data patterns. Even while our project did not require complex heatmaps, understanding their applications was valuable. Complex heatmaps were commonly used in bioinformatics and are visualized using R. The above mentioned literature highlighted the widespread use of R libraries, such as [ggplot2](#) and [pheatmap](#), and functions like [aheatmap\(\)](#), for various purposes.

Specifically, we learned how heatmaps can be integrated with K-Means Clustering to illustrate slice-level hierarchical relationships, effectively displaying the hierarchical clustering performed within each heatmap slice [2].

2.3 Machine learning and dengue forecasting: Comparing random forests and artificial neural networks for predicting dengue burden at national and sub-national scales in Colombia

In order to predict mosquito-borne disease outbreaks up to 12 weeks ahead of time, the study compared Artificial Neural Network (ANN) and Random Forest models. The Random Forest model provided better accuracy and a lower mean error, the authors concluded. We chose Random Forest as one of our models because their paper's objective closely matched our own. Additionally, they used 10-fold cross-validation in their training methodology.

This literature guided us to try and implement Random Forest in our problem statement. It was informative to know how a simple methodology like Random Forest can be used in cases of such big data on a national level [3].

2.4 Optimized Risk Scores

Risk scores are simple classification models that let users assess the threat by performing small to basic operations on the numbers or attributes of the dataset. But coming up with the said formula to calculate the risk score for all problem statements is hard since complex operations cannot just rely on basic arithmetic operations. Therefore, the scholars in this literature ambitiously built a new system for Risk Score calculations and they coined their method as *Risk-calibrated Supersparse Linear Integer Model (RiskSLIM)*. This can be used in models that are highly optimized for feature selection and small integer coefficients.

Their overall goal was to come up with a universal Risk Score calculation approach for highly optimized models. Even though we won't be using RiskSLIM on our models, it was educational to learn about it. This paper was deeply enlightening as it showcases how complex and how dedicated one must be in order to develop their own approach to solve an issue [4].

2.5 Exploring the Power of eXtreme Gradient Boosting Algorithm in Machine Learning: A Review

This paper contains all about machine learning models for a beginner, it helped us explore and decide on which type of models can be applied, but it goes in depth of XGBoost (eXtreme Gradient Boosting). Machine learning models are split into supervised, unsupervised, Neural Networks & Deep Learning, Ensemble. XGBoost falls into the category of an ensemble model. An Ensemble model is specifically used to reduce variances. It trains on multiple datasets which are derived from the main dataset and by averaging the outputs, it reduces the noise or any outliers that are skewing the data. The process is split into two parts called bagging and boosting. Bagging decreases variance while boosting reduces both variance and classification bias.

XGBoost is based on decision trees and it utilizes gradient boosting methods. A key advantage of XGBoost is that it can handle missing values by assigning them an average value, thereby mitigating potential errors. We leveraged XGBoost specifically for its ability to prune and manage missing data effectively. Overall, it is designed to specialize in balancing the model to avoid both overfitting and underfitting [5].

2.6 Emerging and Neglected Infectious Diseases: Insights, Advances, and Challenges

This research paper is based on the growing global concern of infectious diseases, especially those which are new and not commonly discovered. The paper describes how over the past few decades, new infectious diseases have been rising globally, and most of them have come from animals. It also tells us how these diseases are spreading and reappearing due to several factors such as changes in the environment, population increases, and human interaction with animals, thereby becoming a major cause of concern for global stability.

The paper also discusses the major reasons for infectious diseases and how these diseases are being transmitted and affecting the population globally. Many factors relating to human behaviour and activities, pathogen evolution, poverty, and changes in the environment as well as dynamic human interactions with animals have been found to contribute to infectious disease emergence and transmission. It describes how infectious diseases are causing a major number of

deaths globally and how these diseases are spreading by different means. The paper also highlights the significance of health care and global awareness about infectious diseases to prevent them effectively [6].

2.7 Identification of significant climatic risk factors and machine learning models in dengue outbreak prediction

The research paper focuses on how the prediction of dengue fever can be improved by studying how different climatic factors contribute to the spread of dengue fever. It explains that dengue is a major global health concern, with millions of cases reported every year and no fully effective treatment or vaccine available. Because of this, early prediction of outbreaks is very important. The researchers used correlation analysis to identify the most important climate factors and applied multiple machine learning models such as Bayes Network, SVM, RBF Tree, Decision Table, and Naive Bayes on real data collected in Malaysia from 2010 to 2013.

The major contribution of the paper is the introduction of a new combined climatic factor called the TempeRain Factor (TRF), which helped improve prediction accuracy. When tested using machine learning models, especially the Bayes Network model, the use of TRF increased the prediction accuracy to about 92.35%, which is higher than many existing models [7].

2.8 A machine learning-based universal outbreak risk prediction tool

In order to prevent and control the growing rate of serious epidemics, the ability to predict the risk caused by emerging outbreaks is a vital skill. However, most risk prediction tools, apart from EPIRISK, are not capable of carrying out risk prediction due to the limitation of targeting one particular disease and one particular country. To overcome this problem, a risk prediction system based on five machine learning models, namely Neural Network XGBoost, Logistic Boost, Random Forest, and Kernel SVM, was developed, which can predict and give a collective output for a particular problem, and can give accurate predictions with 80% to 90% accuracy. Three different datasets were developed to test and train different ML models under different conditions and situations, and this prediction system has a powerful ability to predict, is adaptable, and can give a universal risk prediction output, which does not depend on geographical boundaries and types of diseases [8].

2.9 Machine Learning and Prediction of Infectious Diseases: A Systematic Review

The research paper aims to assess how machine learning can be employed to predict infectious diseases at an early stage. The research highlights how infectious diseases are a serious problem in the world, causing millions of deaths every year, and how it is important to understand how they are transmitted and what factors influence them. It also highlights how machine learning can play a critical role in analyzing large and complex data sets, identifying patterns, and even predicting trends related to infectious diseases based on different factors such as population, environmental, and behavioral conditions.

It uses machine learning models such as decision trees, random forests, support vector machines, and even deep learning models which can be employed to predict infectious diseases with a high level of accuracy. The research has also highlighted how different machine learning models can be combined to improve predictive accuracy, especially in relation to infectious diseases, and how this can be a critical factor in helping governments and healthcare organizations take preventive actions at an early stage [9].

2.10 Comparing different supervised machine learning algorithms for disease prediction

The research paper focuses on comparing different supervised machine learning algorithms used for disease prediction in the healthcare domain. The research paper also explains how machine learning models are trained on labelled medical data, to predict whether a person has a risk of developing a particular disease or not. The research study reviewed 48 research papers, in which different machine learning models were applied to predict a particular disease, and a comparative study was made on those models, allowing a fair comparison among them. The research study also highlights different supervised machine learning models, such as Support Vector Machine (SVM), Naive Bayes, Decision Tree, Random Forest, Logistic Regression, K-Nearest Neighbor, and Artificial Neural Networks, and how they learn from data and classify a patient into different categories, such as high-risk and low-risk patients, etc.

The research study concluded that even though SVM is one of the most commonly used models, Random Forest gives better accuracy compared to other models. The research study also

highlights how the accuracy of a model can be calculated using different metrics, such as accuracy, precision, recall, and ROC curves, etc [10].

2.11 Machine learning based regional epidemic transmission risks precaution in digital society

According to this paper, "Machine learning based regional epidemic transmission risks precaution in digital society," the authors developed a machine learning model primarily based on an Artificial Neural Network(ANN) to predict regional epidemic transmission risks. This model was specifically designed to address the challenges posed by highly imbalanced datasets, where the occurrence of extreme events is rare.

The central component of their methodology is the Artificial Neural Network. This network is used to learn and identify individuals with higher potential infectious risk and their probability. The basic framework of ANN is fully connected. The ANN incorporates a weighted cross entropy loss function during multi classification training.

The initial results from the neural network are normalized using the Software function. This normalisation provides the probability being designated as the model's prediction [11].

2.12 COVID-19 pandemic hazard–risk–vulnerability analysis: a framework for an effective Pan-India response

This paper focuses on COVID-19 in India. COVID-19, caused by SARS-CoV-2, became a global pandemic in 2020. Marginalized groups and those in certain occupations are more prone to infection. Early lockdown in India had a big impact on vulnerable populations. Despite a high infection rate, India has a relatively low mortality rate so far. Identifying COVID-19 hazard and socio - economic vulnerability is important for predicting health burden and formulating control strategies.

The study uses the fuzzy analytical hierarchical process (F-AHP) and geospatial modeling to evaluate the COVID-19 risk. It uses the number of COVID-19 positive cases and death tolls in India as hazards. Developed states are more severely affected due to higher mobility and

population density. The lockdown in India was effective as the infection rate increased after the unlock phases. Recovery trends are similar to the trends of confirmed cases. Large urban centers and megacities are hotspots of the pandemic. COVID-19 hazards are high in some parts of India, mainly in southern and south-western regions. A large part of the country has moderate hazards, and central, northern and northeastern India have low to very low hazards.

This study defines risk based on population attributes and the number of positive cases. However, due to a lack of data, factors such as the number of hospital facilities, housing density, access to hospitals, personal immunity, interpersonal interactions, human behavior and virus exposure are not included in the study. The study has identified the role of geography in the COVID-19 outbreak in India and major hotspots. The study provides a method for monitoring the progression of COVID-19 outbreaks, which can contribute to strategic planning and resource allocation to reduce the impact of the pandemic [12].

2.13 Assessing Socioeconomic Vulnerabilities Related to COVID-19 Risk in India: A State-Level Analysis

The COVID-19 pandemic has been a global catastrophe. In India, it has also had a significant impact. Given India's large and dense population, aging population, poor healthcare system, regional inequalities, and high population mobility, it is at greater risk of increasing COVID-19 infections.

Data for the study was compiled from multiple sources. Descriptive statistics was used to analyse the data. Principal Component Analysis (PCA) was used to examine the predictors of COVID-19 risk. The confirmed positive COVID-19 cases were used as the main outcome variable. Bivariate and multivariate techniques were applied. A negative binomial regression model was used to identify the association of various factors with the COVID-19 pandemic. Principal component analysis was used to compute the 'health infrastructure index'.

There were substantial inter-state differentials in the prevalence rate of COVID-19 positive cases in India from March 16, 2020, to May 3, 2020. The average weekly new confirmed cases increased continuously. The recovery rate improved. The average weekly deaths due to

COVID-19 also increased. Demographic, socioeconomic, and public health resilience factors were analyzed in relation to COVID-19 positive cases. A negative binomial regression model showed various factors associated with COVID-19 risk. The composite vulnerability index suggested that eastern and parts of northern states were highly vulnerable.

2.14 A rapid risk analysis tool to prioritise response to infectious disease outbreaks

A risk analysis framework was developed by modifying an Ebola specific tool. They developed a risk analysis tool called EpiRisk, it used simple linear summation model. It includes both disease specific and country specific parameters. Initial factors were categorized as disease related or country related. Country related data were collected from sources like the World Bank, and disease related data from grey literature and official sources. Values from 1 to 3 were assigned to each parameter based on risk level. After qualitative and quantitative analyses, 14 parameters were used. An outbreak in a country is given scores in disease and country factors, which are summed for a final risk score. The framework was evaluated using past outbreaks and data on 61 outbreaks between 2002 - 2018.

As a result, for the nine historical outbreaks analyzed, Australia had the lowest overall risk score and Nigeria had the highest total country score. For the 61 - outbreak evaluation dataset, the risk scores were split into quantiles, as were the number of deaths. There was a clear association between the risk scores computed by the framework and the severity of outbreaks as measured by deaths, but some outliers were present. For example, some severe outbreaks had low risk scores and vice versa [14].

2.15 Epigenetic scores of blood-based proteins as biomarkers of general cognitive function and brain health

The global prevalence of dementia is projected to reach 152 million by 2050. Dementia is characterized by cognitive decline that affects daily activities and quality of life. Biomarkers could facilitate early detection and intervention. Proteins are commonly used as biomarkers, and blood-based biomarkers are desirable due to their accessibility and cost-effectiveness. It used linear models, Logistic regression models and Cox proportional hazards model.

Cognitive testing, MRI measures, and dementia diagnosis data were collected. Statistical analyses included regression models, meta-analyses, and gene ontology analysis. Several EpiScores were significantly associated with general cognitive function in the basic and fully adjusted models, and a meta-analysis identified 18 significant EpiScores. No significant associations were found between EpiScores and cognitive change. Some EpiScores were associated with MRI markers of brain health, and a few were associated with a binary dementia diagnosis and time-to-dementia. However, no biological pathways were significantly enriched. The study identified multiple associations between protein EpiScores and measures of cognitive function, brain health, and dementia. The lack of replication in some results could be due to using all-cause dementia as a phenotype.

The paper concludes that 84 protein EpiScores were tested against measures of general cognitive function, brain health, and incident dementia across three human cohorts. Several EpiScores may augment typical risk factors of brain health, but further replication studies are needed [15].

2.16 Predicting Infectious Disease Outbreaks with Machine Learning and Epidemiological Data

This study proposes how machine learning can be used to predict infectious disease outbreaks more effectively. Instead of depending only on traditional methods, it combines machine learning with epidemiological data to understand how diseases spread. The data used in such studies includes information like number of cases, deaths, location, and time. Along with this, environmental factors such as temperature and rainfall are also considered. Most of the disease data is collected from health surveillance systems which regularly report outbreaks of diseases like dengue and malaria. The steps followed in the paper starts with collecting and cleaning the data. Missing values and errors are handled. Then, important features are selected to improve the model's performance. After that, machine learning models are applied to learn patterns from past data and predict future outbreaks.

This work also explains different types of predictions such as time-based prediction, location-based prediction, and risk prediction. This helps in identifying when and where

outbreaks might happen and which areas are more at risk. The main idea of the paper is that combining machine learning with real-world disease data gives better and more accurate results. It helps in early detection and can support authorities in taking preventive actions. However, some challenges like incomplete data and lack of real-time information are also mentioned. Still, the study shows that machine learning is very useful in improving disease prediction [16].

2.17 Machine Learning for Infectious Disease Risk Prediction: A Survey

The paper mainly focuses on different stages of disease spread, such as epidemic and endemic phases. Firstly, the goal is to predict how fast a disease will spread later, the focus is on controlling and reducing the spread. Machine learning models are designed differently based on these stages. Different types of models are explained in the paper. Traditional models like regression and statistical methods are used for basic predictions. More advanced models like RNN, CNN, GNN are used to capture complex patterns such as time-based trends and geographical spread. These models help in understanding how diseases move across regions and over time. The paper also explains that combining machine learning with epidemiological models like SIR and SEIR gives better results.

While machine learning provides accuracy, epidemiological models give real-world meaning to the predictions. So, combining both helps in making better decisions for public health. However, there are some challenges mentioned, like lack of proper data, missing values, and difficulty in evaluating models. Overall, this paper shows that machine learning plays a very important role in predicting disease outbreaks and helping governments take early action [17].

2.18 Climate Change Impacts on the Global Spread of Malaria and Dengue

This paper studies how climate change is affecting the spread of diseases like malaria and dengue across the world. It explains that changes in temperature, rainfall, and weather conditions directly impact how these diseases spread. The study is based on a review of multiple research papers. The authors collected and analyzed data from different regions to understand how climate change is affecting disease patterns. They found that warmer temperatures and changes in rainfall create suitable conditions for mosquitoes, which increases the spread of these diseases.

The paper also explains how different regions are affected differently. For example, malaria is more affected in African regions, while dengue is more common in Southeast Asia. It also shows that diseases are slowly spreading to new areas due to climate change. Another important part of the paper is about adaptation strategies. It suggests methods like better surveillance systems, early warning systems, and improved healthcare planning to reduce the impact of these diseases. Still, the paper also mentions that many regions lack proper implementation of these strategies. Overall, the study concludes that climate change has a strong impact on disease spread, and proper planning and early prediction systems are necessary to control future outbreaks [18].

2.19 Mapping Malaria Risk in India Between 2019–2023

This paper focuses on analyzing how malaria has spread across different districts in India over the years 2019 to 2023. The main goal is to identify high-risk areas and understand the factors that influence malaria transmission. The researchers used district-level malaria data collected from government sources along with climate data such as temperature, rainfall, and humidity. They also included land-use and socio-economic data to get a more complete understanding. By combining all these datasets, they were able to perform a detailed analysis of malaria patterns across India. One important method used in this paper is spatial analysis, where maps are used to show how malaria cases are distributed across different regions. The study also used statistical techniques to identify hotspot areas where malaria cases are highly concentrated. These hotspots were mainly found in central and north-eastern regions of India.

The paper also applies machine learning models such as Random Forest and Zero-Inflated Poisson Regression to identify the key factors affecting malaria spread. These models help in understanding which variables like climate or socio-economic conditions have the strongest impact on disease transmission. The results show that malaria cases decreased during the COVID-19 period but started increasing again afterward. It was also observed that some regions consistently remained high-risk, while some areas showed changes over time. Another important aspect is that vulnerable populations, especially economically weaker groups, are more affected by malaria. Additionally, the study developed an interactive dashboard that allows users to visualize malaria data and track changes over time. This makes the research more practical and

useful for public health planning. Overall, this paper highlights the importance of combining data analysis, machine learning, and visualization techniques to better understand and control disease outbreaks [19].

3.20 Machine Learning-Enabled Multi-Hazards Risk Assessment Considering Social Vulnerability

This paper focuses on assessing risk by combining multiple natural hazards with social vulnerability using machine learning techniques. Instead of looking at hazards only, the study emphasizes that risk also depends on how important the population is. The authors considered three major hazards: flooding, wildfires, and seismic activity. Along with this, they also analyzed social factors such as population, income, education, and infrastructure, which influence how severely a community is affected by these hazards. The methodology involves collecting data from different sources and performing preprocessing steps like cleaning, scaling, and feature selection. The dataset is then divided into training and testing sets to build and evaluate machine learning models.

Several machine learning models are used in this study, including Naive Bayes, K-Nearest Neighbors, Logistic Regression, Random Forest, and K-Means clustering. Among these, Random Forest was found to give the best performance in predicting both hazard levels and social vulnerability. The paper also introduces the concept of combining hazard levels with vulnerability levels to calculate overall risk. The results are presented using maps that show different regions classified into low, medium, and high-risk categories. These visualizations make it easier to identify critical areas that require immediate attention. Overall, the study concludes that considering social vulnerability along with hazards provides a more accurate understanding of risk [20].

Chapter 3

Problem Statement

Despite clear links between climate and epidemic spread, India lacks scalable, non-GIS methods for early identification of high-risk regions. Existing systems are infrastructure heavy, hard to implement and difficult to learn. This creates a gap in identifying high epidemic-prone risk zones for preventative actions. The approach should be scalable and data driven that can quantify and visualize the epidemic zones district wise using readily available data. This allows us to assist and allocate the high risk zones with healthcare and infrastructure to reduce impacts or outbreaks.

Chapter 4

Objectives

The primary objective of this project is to build a scalable, data driven framework that can identify and rank high risk epidemic zones across Indian districts, using publicly available climate and disease incidence data. The specific objectives are outlined below.

4.1 Data Acquisition and Preprocessing

The first objective involves collecting district-level epidemic and climate data from reliable open-access sources including the EpiClim dataset and the Indian Meteorological Department (IMD). The collected raw data was cleaned to remove duplicates and inconsistencies, and handle missing values. Exploratory Data Analysis (EDA) will be performed to understand the distribution of cases, deaths, and climatic variables across districts and years.

4.2 Risk Score Computation

The second objective is to design and compute a composite epidemic risk index for each district. The risk score will be formulated by assigning weighted contributions to key climatic and epidemiological variables, including rainfall, temperature and Leaf Area Index (LAI). Districts will then be ranked and categorized into high, moderate, and low-risk zones based on their computed scores, enabling a straightforward, interpretable classification without the need for GIS software.

4.3 Clustering-Based Zone Identification

The third objective is to apply unsupervised machine learning clustering algorithms to group districts that exhibit similar epidemic risk profiles. K-Means clustering will be the primary method, supplemented by DBSCAN for detecting denser outbreak hotspots. This step allows the model to discover natural groupings within the data, independent of manually defined thresholds, thereby lending greater objectivity to the risk zone classification.

4.4 Supervised Classification for Risk Prediction

The fourth objective involves training supervised classification models to validate and extend the clustering results. A Random Forest classifier and an XGBoost model will be trained on features derived from the risk scoring and clustering phases, with the goal of predicting whether a given district falls into a high-risk or low-risk category. These models will be evaluated using standard metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to ensure robustness and generalizability.

4.5 Visualization and Reporting

The final objective is to communicate the findings in a clear and accessible manner through heatmaps, bar charts, and time-series plots. These visualizations will depict how epidemic risk varies by state, district, season, and year. The insights derived will be translated into actionable policy recommendations that can help public health authorities prioritize surveillance efforts, pre-position medical resources, and implement preventive interventions in the highest-risk zones.

Chapter 5

Dataset Description

5.1 Overview of Datasets Used

This project integrates two primary datasets: the EpiClim dataset, which provides district-level weekly epidemic and climate records for India, and supplementary meteorological data from the Indian Meteorological Department (IMD). Together, these datasets offer a comprehensive view of how disease incidence and climatic conditions have co-evolved across Indian districts over more than a decade.

5.2 EpiClim Dataset

The EpiClim dataset is a publicly accessible, peer-reviewed compilation hosted on arXiv [1]. It was assembled to support geo-health research by bringing together outbreak records from India's Integrated Disease Surveillance Programme (IDSP) and co-located climate variables derived from satellite and meteorological sources. The IDSP publishes weekly outbreak alerts across Indian states and union territories, and EpiClim standardizes these reports into a structured, machine-readable format.

The dataset spans from 2009 to 2022, offering 14 years of longitudinal data. It covers 36 states and union territories of India, and includes records from 795 unique districts. Each record is geo-located with the latitude and longitude coordinates of the corresponding district centroid, enabling spatial analysis even in the absence of dedicated GIS tools.

The consolidated dataset used in this project, referred to as Final_data.xlsx, contains 8,985 records across 15 columns. Table 5.1 describes each feature along with its data type and role in the analysis.

Table 5.1: Description of features

Feature	Data Type	Unit / Range	Description
week_of_outbreak	String	1st–52nd week	Week of the year in which the outbreak was recorded
year	Integer	2009–2022	Calendar year of the outbreak record
state_ut	String	36 states/UTs	Name of the Indian state or union territory
district	String	795 unique	District where the outbreak was reported
Disease	Categorical	22 categories	Type of infectious disease (e.g., Dengue, Malaria, Cholera)
Cases	Integer	≥ 0	Number of confirmed disease cases in the week
Deaths	Integer	≥ 0 / NaN	Number of deaths; missing in approximately 6,431 records
day	Integer	1–31	Day of month corresponding to outbreak date
mon	Integer	1–12	Month of outbreak (1 = January, 12 = December)
Latitude	Float	8°N – 37°N	Latitude of the district centroid
Longitude	Float	68°E – 97°E	Longitude of the district centroid
preci	Float	0.00–5.68 mm	Average weekly precipitation in the district
LAI	Float	0–35+	Leaf Area Index, a proxy for vegetation density and land cover
Temp	Float	259.68–327.73 K	Mean weekly surface temperature (in Kelvin)

The dataset captures 22 distinct disease categories, though many are subcategories or alternate spellings of core diseases. The primary disease categories and their record counts are summarized in Table 5.2. Acute Diarrhoeal Disease dominates the dataset, followed by Dengue and Chikungunya. Across all records, the dataset documents a total of 796,427 cases and 4,047 reported deaths over the 14-year period. The figure 5.1 just helps us visualize the overall distribution of diseases.

Distribution of diseases

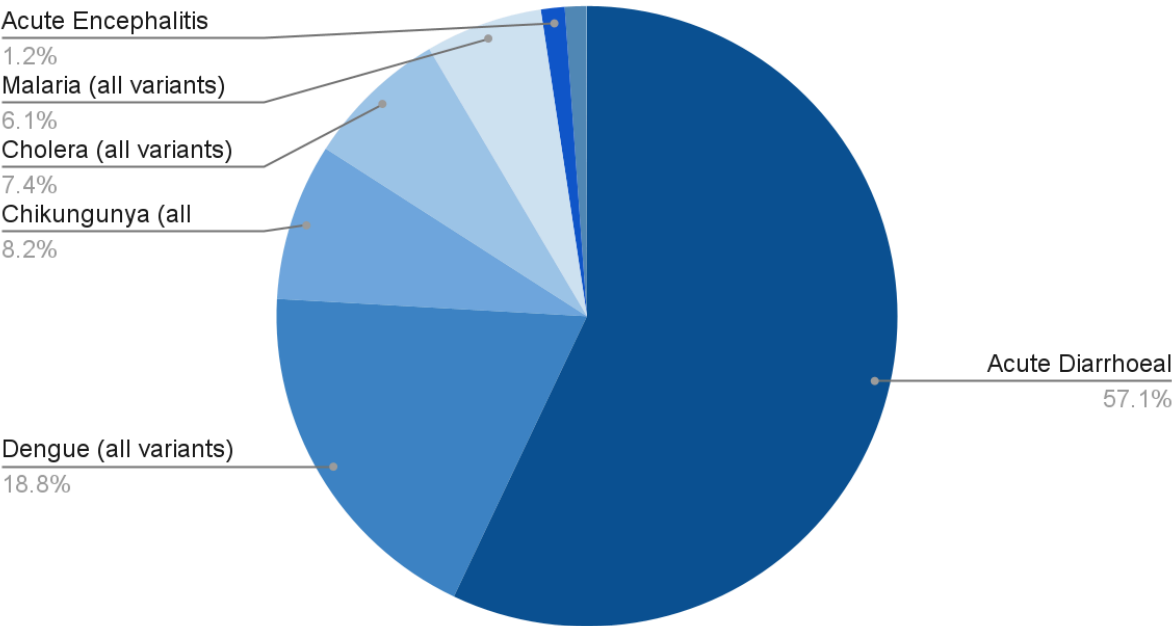


Figure 5.1: Pie chart for distribution of diseases

Table 5.2: Distribution of disease records in the dataset

Disease Category	Number of Records
Acute Diarrhoeal Disease	5,126
Dengue (all variants)	1,693
Chikungunya (all variants)	736
Cholera (all variants)	668
Malaria (all variants)	548
Acute Encephalitis Syndrome	111
Acute Gastroenteritis	100
Other/Mixed	3

5.3 IMD Rainfall Dataset

In addition to the EpiClim dataset, district-level rainfall data published by the Indian Meteorological Department (IMD) was incorporated into the analysis. The IMD dataset provides monthly rainfall figures across Indian districts and serves as an independent verification of the precipitation variable already present in the EpiClim records. Cross-referencing both sources helped improve data quality and fill gaps where EpiClim records were missing precipitation values.

Chapter 6

Methodology

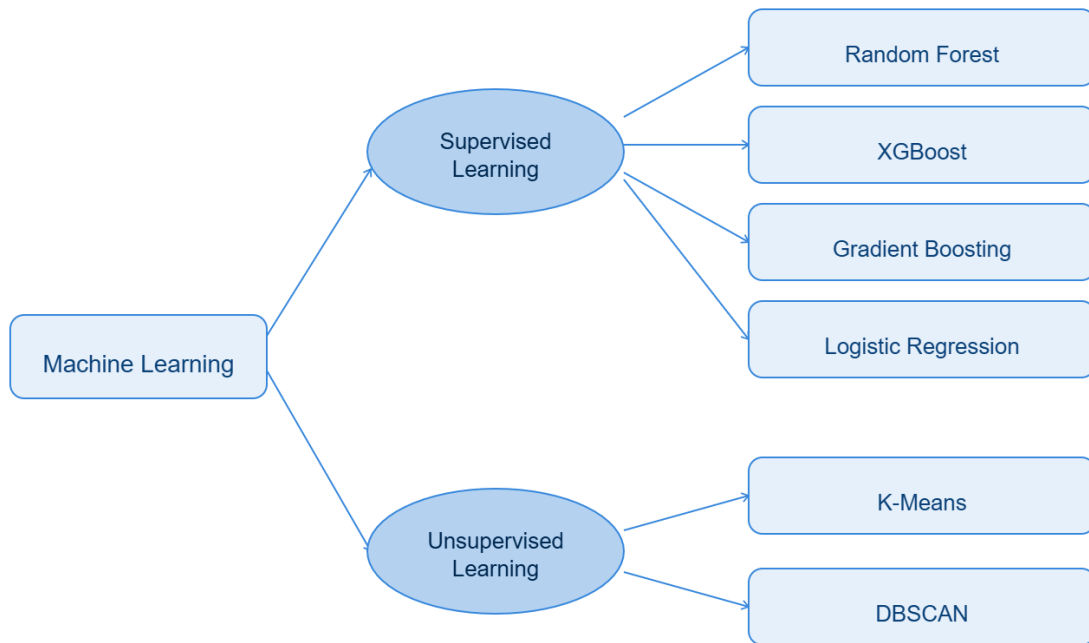


Figure 6.1 Methodology process

The methodology followed in this project consists of five major stages: data preprocessing, risk score computation, unsupervised clustering, supervised classification, and visualization. Each stage builds upon the results of the previous, and together they form a coherent end-to-end pipeline for identifying and ranking epidemic risk zones across Indian districts. The figure 6.1 showcases the approach of machine learning and also shows us the models used. It visualizes the type of model used for the type of data.

6.1 Data Preprocessing

The dataset was preprocessed to ensure consistency and accuracy for machine learning analysis. First, irrelevant columns like “Unnamed: 0” were removed. Numerical columns including Cases, Temperature, Deaths, Month, Latitude, Longitude, Precipitation, and LAI were changed to proper numeric formats to address inconsistencies in the dataset. Missing values were dealt with using mean imputation for key features like Cases, Temperature, Precipitation, and LAI to maintain data flow. To improve data quality, disease names were standardized by changing them to lowercase and combining similar categories like “suspected dengue” and “dengue fever” into one. This reduced redundancy and improved analysis accuracy. Also, a new categorical feature called “Type” was added to classify diseases into two groups: Vector-borne diseases (like dengue, malaria, and chikungunya) and Water-borne diseases (like diarrheal diseases). This allowed for comparison between disease types. Overall, preprocessing created a clean, structured dataset ready for further analysis and modeling.

6.1.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) helped us understand patterns, trends, and relationships in the dataset. We used different visualizations to look at disease distribution, seasonal trends, correlations, and high-risk areas. Through EDA, we identified important insights like seasonal outbreak patterns, main diseases, and geographic risk zones. These findings guided our choice of features and the development of our model.

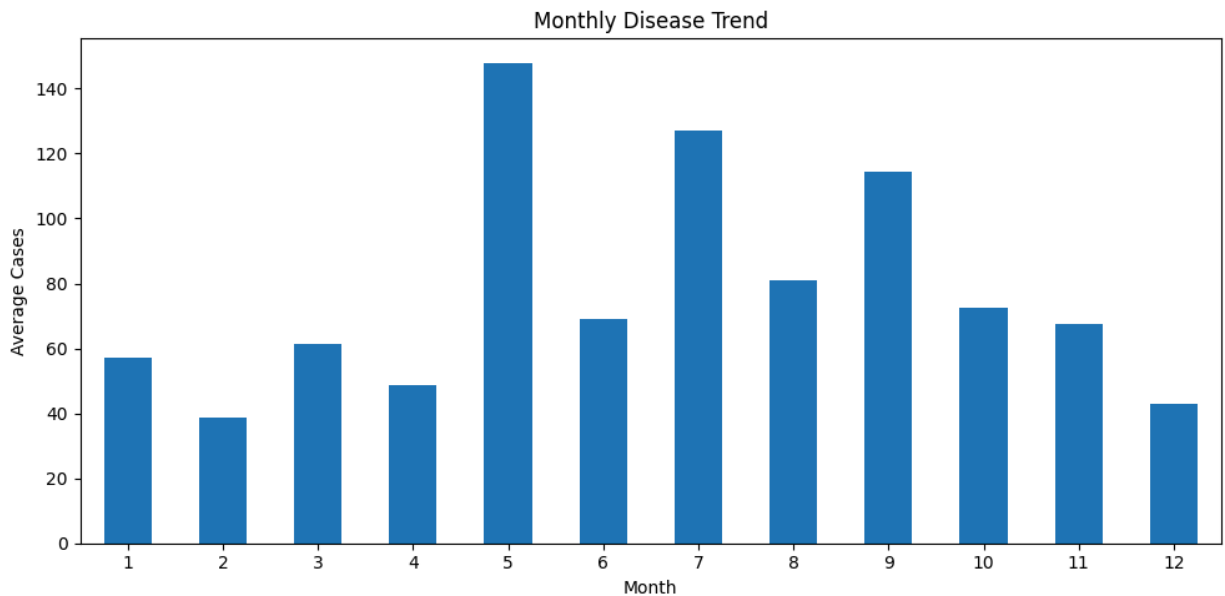


Figure 6.1.1: Monthly Disease Trend

Figure 6.1.1 shows the monthly distribution of average disease cases. Cases peak from months 5 to 9, with the highest number in month 5. This timeframe aligns with the monsoon season, suggesting that weather factors like rainfall and humidity play a role in spreading the disease. This pattern emphasizes how seasonal patterns are important for predicting epidemics.

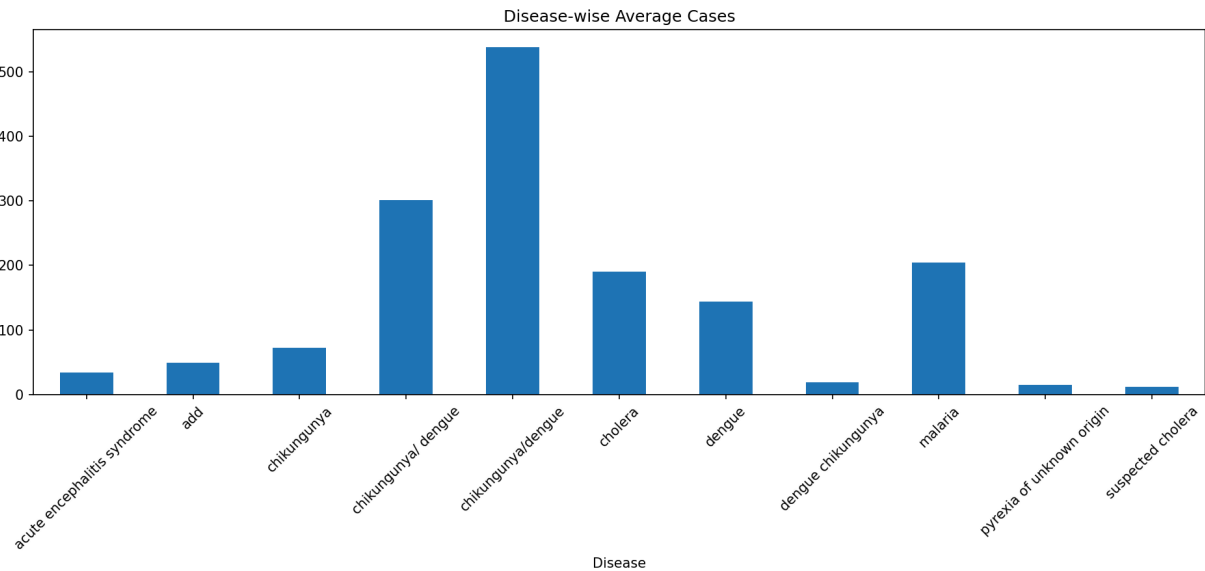


Figure 6.1.2: Disease-wise Average Cases

Figure 6.1.2 shows the average number of cases across different diseases. It can be observed that diseases such as dengue and chikungunya have significantly higher case counts compared to others, while diarrhoeal diseases also show considerable prevalence. This indicates that vector-borne diseases contribute more to the overall disease burden in the dataset. The analysis highlights the importance of focusing on these diseases for effective prediction and prevention strategies.

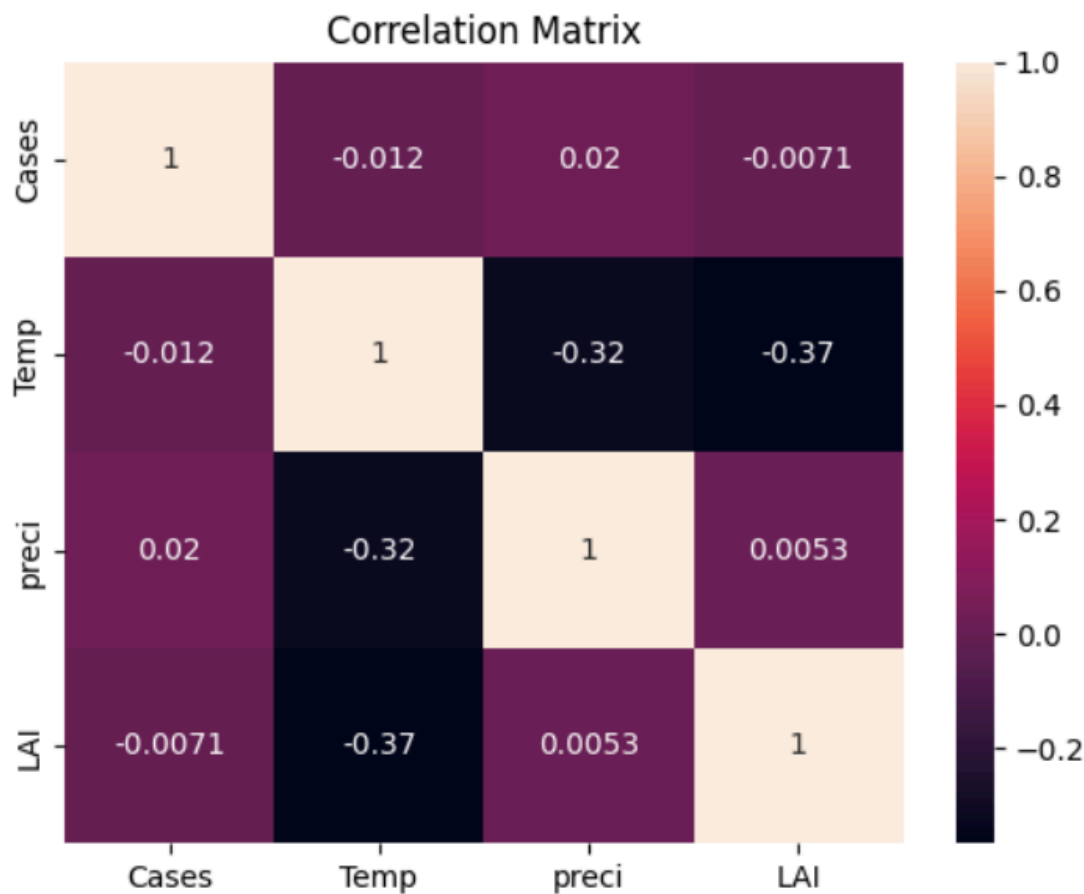


Figure 6.1.3: Correlation Matrix

Figure 6.1.3 presents the correlation matrix between key variables such as Cases, Temperature, Precipitation, and LAI. It can be observed that the correlation between cases and climatic variables is relatively weak, indicating that disease spread is influenced by multiple factors rather than a single variable. Moderate negative correlations can be seen between temperature and other features, suggesting indirect relationships. This highlights the complexity of disease prediction and justifies the use of machine learning models to capture non-linear patterns in the data.

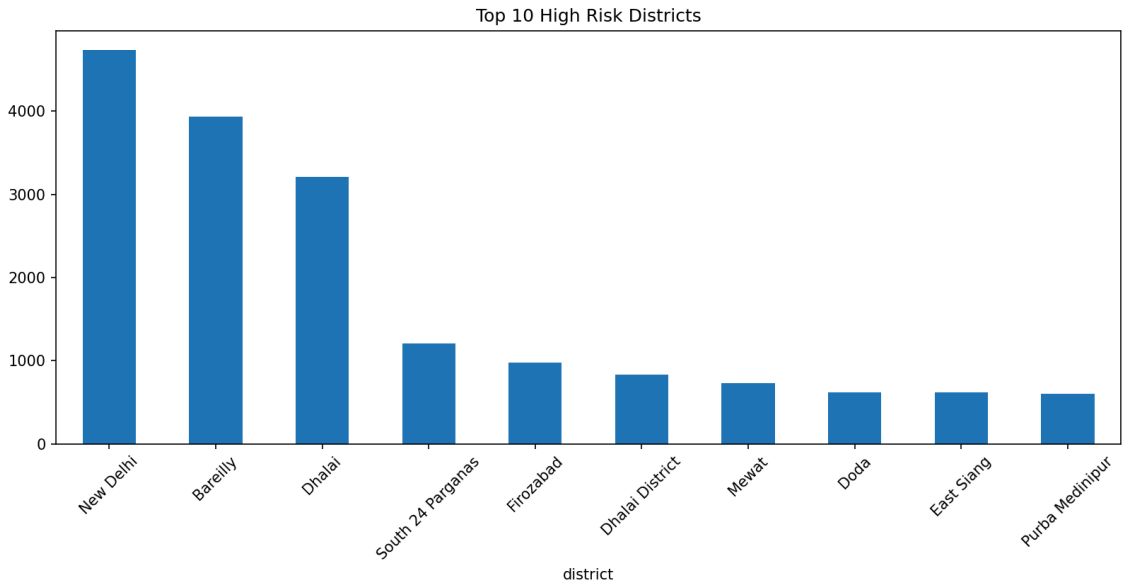


Figure 6.1.4: Top 10 High-Risk Districts

Figure 6.1.4 shows the top 10 districts with the highest average number of disease cases. It can be observed that districts such as New Delhi, Bareilly, and Dhalai have significantly higher case counts compared to others. This indicates that certain regions consistently experience a higher disease burden, possibly due to factors such as population density, urbanization, environmental conditions, and healthcare accessibility. Identifying these high-risk districts is important for targeted interventions and efficient allocation of healthcare resources.

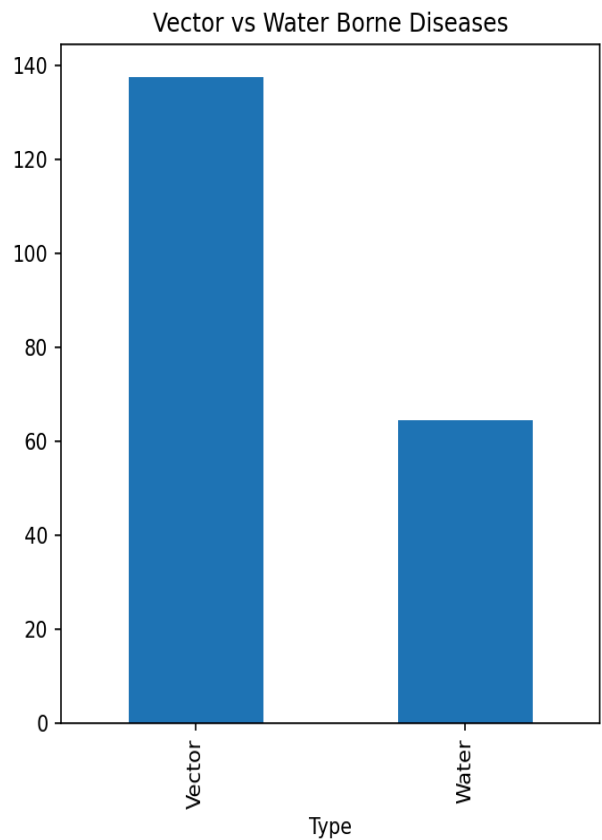


Figure 6.1.5: Vector vs Water-Borne Diseases

Figure 6.1.5 compares the average number of cases between vector-borne and water-borne diseases. It can be observed that vector-borne diseases have significantly higher case counts compared to water-borne diseases. This indicates that diseases such as dengue, malaria, and chikungunya are the dominant contributors to the overall disease burden in the dataset. This insight highlights the importance of focusing on vector control and environmental factors in disease prevention and prediction.

6.2 Risk Score Computation

A Risk Score is a composite numerical value that represents the overall level of risk associated with a district or region by combining multiple influencing factors into a single measurable index. The risk index is made up of a series of weighted factors, all of which contribute to the

calculation of the epidemic risk levels that exist in each district along a numerical scale. Such an index is used for calculating the probabilities and consequences associated with negative events such as epidemics, environmental impacts, and climatic events.

The formula for a weighted risk score using normalized variables is the sum of each variable's normalized value multiplied by its assigned weight, divided by the sum of all weights.

$$\text{Risk Score} = \frac{w1 \times \text{Cases} + w2 \times \text{Rainfall} + w3 \times \text{Temperature} + w4 \times \text{LAI}}{w1 + w2 + w3 + w4}$$

where:

- $w1$, $w2$, $w3$ and $w4$ are the weights assigned to each variable
- All variables are normalized to ensure comparability

Risk categorization was performed using a quantile-based approach, dividing the normalized risk scores into three equal groups (tertiles) representing low, moderate, and high risk. This method allows relative comparison of districts based on their risk levels and ensures balanced classification.

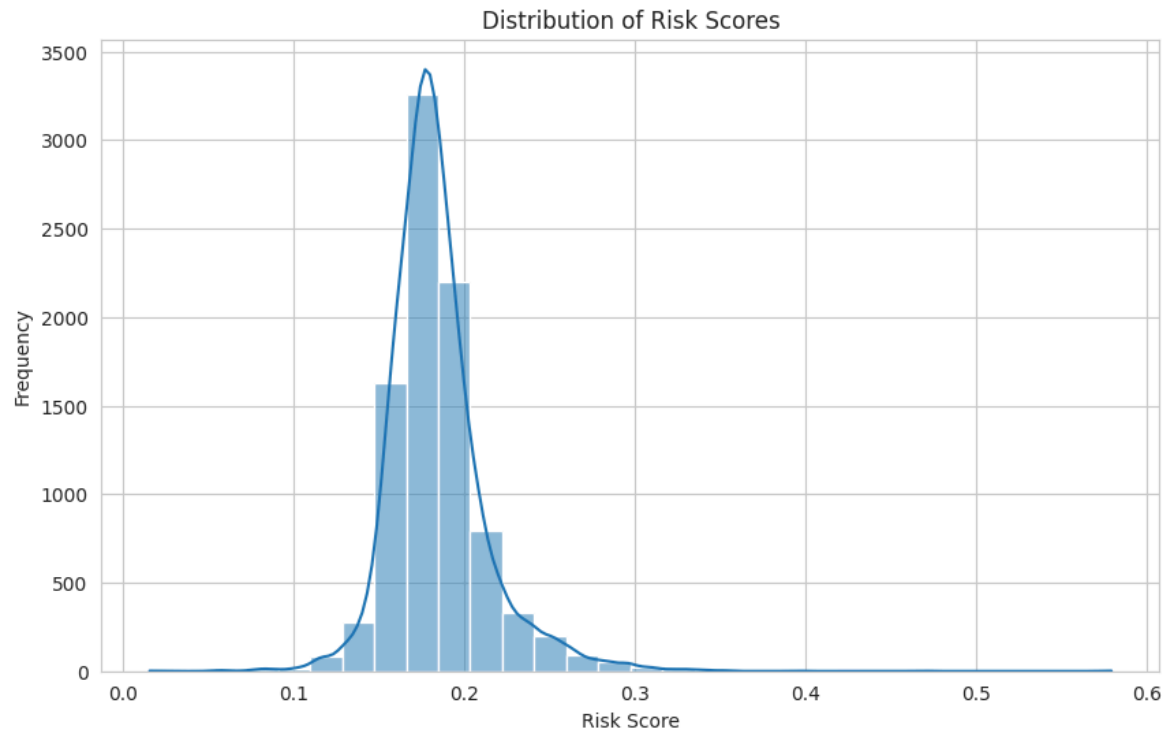


Figure 6.2.1 Distribution of Risk Scores

Figure 6.2.1 shows the distribution of risk scores using a histogram with a Kernel Density Curve. It can be observed that most of the risk values belong to the moderate risk level, forming a peak in this range. The distribution shows that most values lie around 0.2, indicating that a large number of regions fall under moderate to high-risk categories based on the defined thresholds. The distribution is slightly right-skewed, with a few regions having higher risk scores above 0.3. This suggests that while most areas experience moderate disease risk, a smaller number of regions are highly vulnerable and require focused attention.

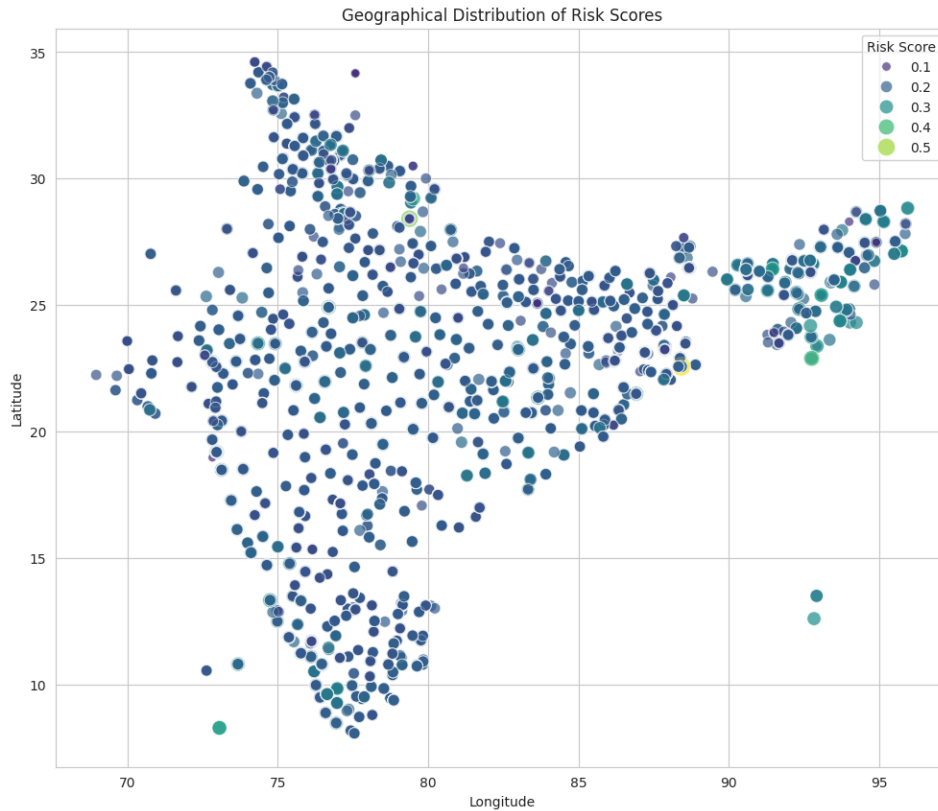


Figure 6.2.2 Geographical Distribution of Risk Scores

Figure 6.2.2 illustrates the geographical distribution of risk scores across different regions using latitude and longitude coordinates. Each point represents a specific location, and the color intensity indicates the level of risk, ranging from low to high. It can be observed that most regions fall under low to moderate risk levels, as indicated by blue and light green points. However, certain clusters of higher risk values (green to yellow) are visible in specific regions, suggesting localized hotspots of disease prevalence.

6.3 Clustering-Based Risk Zone Identification

Unsupervised clustering was applied to the district-level feature to identify naturally occurring groupings among districts with similar epidemic profiles.

6.3.1 K-Means Clustering

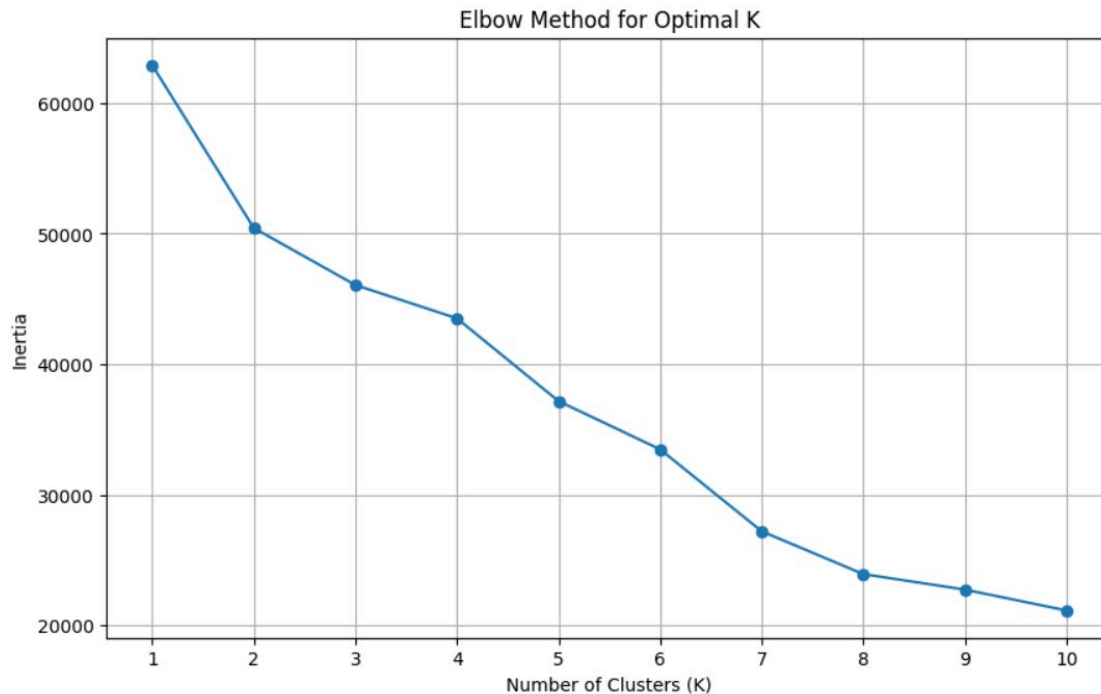


Figure 6.2.3 The Elbow Method for Optimal K

In the Figure 6.2.3, The elbow method is basically a way to figure out how many groups (clusters) to split our data into. Too few groups and you're lumping very different districts together. Too many and you're overcomplicating things for no real reason.

Looking at our graph, when we go from 1 cluster to 2, the inertia drops a lot from around 63,000 all the way down to about 50,000. It keeps dropping as we add more clusters, but the drops get smaller and smaller. Around $K=3$, the curve starts to flatten out and stops dropping as steeply. That point where it kind of bends, that's the "elbow", and that's where we stop.

After $K=3$, adding more clusters gives you less and less benefit. Like going from 8 to 9 clusters barely changes anything at all. So we went with 3 clusters, which also made practical sense for our project — it maps nicely to low risk, moderate risk, and high risk zones. Three groups is simple enough to explain to someone but still meaningful enough to actually be useful.

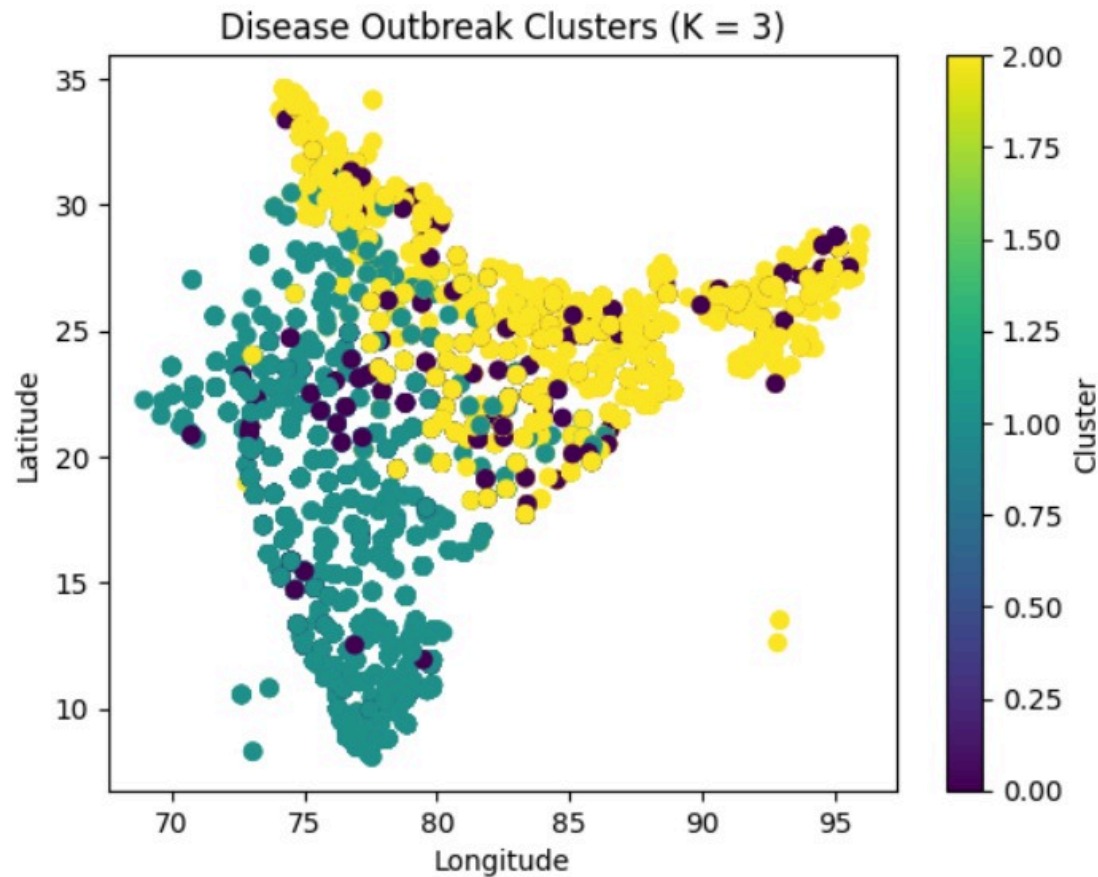


Figure 6.2.4 Disease Outbreak Clusters(K=3)

K-Means clustering (Figure 6.2.4) was applied as the primary clustering algorithm due to its computational efficiency, interpretability, and well-established effectiveness in public health data analysis [3]. The algorithm partitions the 795 districts into k clusters by minimizing the within-cluster sum of squared distances. The optimal number of clusters was determined using the Elbow Method (Figure 6.2.3), which plots the inertia (total within-cluster variance) against increasing values of k . A value of $k=3$ was selected to align with the three-tier risk classification (high, moderate, low).

6.3.2 DBSCAN Clustering

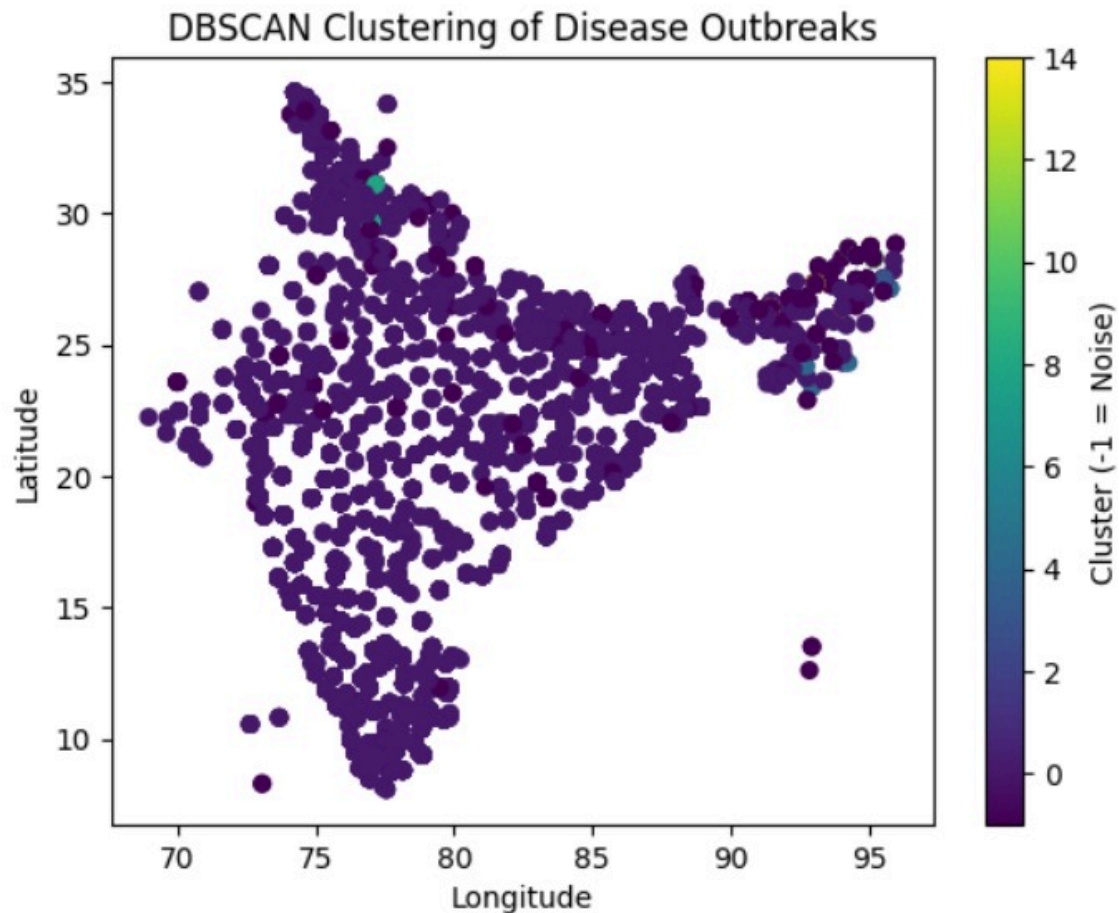


Figure 6.2.5 Disease Outbreak Clusters(K=3)

Density-Based Spatial Clustering (Figure 6.2.5) of Applications with Noise (DBSCAN) was applied as a complementary method to detect high-density outbreak hotspots without requiring a pre-specified number of clusters. Unlike K-Means, DBSCAN is particularly effective at identifying irregularly shaped clusters and flagging outlier districts that do not belong to any well-defined group. These outlier districts often correspond to regions with extreme, sporadic outbreaks that do not follow broader regional patterns, and their identification is valuable for targeted surveillance.

6.3.3 Cluster Quality Evaluation

Cluster quality was evaluated using the Silhouette Score, which measures how similar each district is to its own cluster compared to its nearest neighboring cluster. A silhouette score close to 1 indicates well-separated, internally coherent clusters, while a score close to 0 or negative indicates overlapping or poorly defined clusters. The Davies-Bouldin Index was also computed as a secondary metric; lower values indicate better clustering performance.

6.4 Supervised Classification for Risk Prediction

To provide a predictive layer to the framework, supervised classification models were trained to predict whether a district falls into a high-risk or non-high-risk category based on its climatic and epidemiological features. The cluster labels and risk score tiers from the previous stages were used as the ground truth labels for training.

6.4.1 Random Forest

Random Forest [3] was selected as the primary classification model due to its strong track record in disease prediction tasks, its robustness to outliers and irrelevant features, and its built-in mechanism for ranking feature importance. The model was trained using 10-fold cross-validation to ensure generalizability. Hyperparameters including the number of trees, maximum tree depth, and minimum samples per leaf were tuned using grid search.

6.4.2 XGBoost

XGBoost (eXtreme Gradient Boosting) [5] was used as a complementary model to Random Forest. XGBoost builds an ensemble of decision trees in a sequential, gradient-boosted fashion, where each new tree is trained to correct the errors of the previous ensemble. A notable practical advantage in this project is XGBoost's native ability to handle missing values by learning the optimal imputation direction during training, reducing sensitivity to the data quality issues present in the deaths column.

6.4.3 Logistic Regression (Baseline)

A Logistic Regression model was included as a simple, interpretable baseline. While it is unlikely to outperform the ensemble methods, it provides a lower bound on model performance

and allows for direct inspection of the coefficient weights assigned to each feature, thereby adding transparency to the classification process.

6.4.4 Gradient Boosting

Gradient boosting basically trains a series of models one by one, where each new model looks at what the previous one got wrong and tries to fix it. Instead of one big smart model, you get a team of small, dumb trees where each one is specifically studying the last one's mistakes. Keep doing this — fix errors, add a tree, fix the remaining errors, add another tree — and you end up with something surprisingly accurate. It works great for both predicting numbers (like house prices) and sorting things into categories (like spam vs. not spam), and it especially shines when your data lives in rows and columns rather than images or text.

6.4.5 Evaluation Metrics

All models were evaluated using accuracy, precision, recall, F1-score, and the area under the ROC curve (ROC-AUC). Given the class imbalance between high-risk (25% of districts) and non-high-risk (75%) districts, macro-averaged F1-score and ROC-AUC were treated as the primary metrics rather than accuracy alone. Confusion matrices were also generated to visualize the distribution of true positives, false positives, true negatives, and false negatives for each model.

6.5 Justification for Chosen Techniques

The selection of K-Means and DBSCAN for clustering reflects a balance between scalability and flexibility. K-Means is straightforward to implement, easy to interpret, and scales well to the district-level dataset, while DBSCAN adds the ability to identify geographic outliers without forcing every district into a predefined cluster. Random Forest and XGBoost were chosen for classification because the literature consistently reports superior accuracy for these ensemble methods on health and epidemiological datasets compared to single-tree or linear models [3][8][10]. The combination of unsupervised and supervised methods also ensures that the final risk classification is not entirely dependent on any single algorithmic assumption, making the framework more robust and credible for public health decision-making.

Chapter 7

Implementation

7.1. Data Loading and Preprocessing:

The dataset was loaded using the Pandas library from an Excel file (Final_data.xlsx). The “Cases” column was converted to numeric format using `pandas.to_numeric()` to handle inconsistent or missing values. The dataset was sorted based on district, year, month, and day to maintain proper temporal order. Missing and invalid values were handled using imputation techniques such as mean filling and forward grouping methods.

7.2. Feature Engineering:

Lag features were created using Pandas' `groupby()` and `shift()` functions to help the models understand how outbreaks had been developing over time. Two of these were built, one looking back at the previous record's case count, and another reaching back two steps. A rolling average was also calculated over a 3-record window to smooth out short-term spikes and give a cleaner sense of the overall trend.

Death rate was engineered simply by dividing deaths by cases, with 1 added to the bottom to avoid any division by zero errors. Temperature and precipitation were multiplied together to create a single interaction feature, on the basis that their combined effect on disease spread is greater than either factor alone.

Lastly, disease categories were text labels, so they were converted into numbers using Label Encoding before being fed into the models.

7.3. Target Variable Creation:

A binary classification target variable “Outbreak” was created.

A threshold of 10 cases was used:

$$\text{Outbreak} = 1 \text{ if Cases} > 10$$

Otherwise = 0

This converted the problem into a supervised learning classification task.

7.4. Feature Matrix Preparation:

- A final feature set was created including epidemiological, climatic, spatial, and temporal variables:
 - Deaths, Temperature, Precipitation, LAI
 - Latitude and Longitude
 - Lag features and rolling mean
 - Death Rate and interaction features
 - Encoded disease type
 - Month and day
- Missing values in the feature matrix were handled using mean imputation.
- All features were converted to numeric format for machine learning compatibility.

7.5. Train-Test Strategy (Cross Validation):

- Stratified K-Fold Cross Validation (K = 10) was used to ensure balanced class distribution in each fold.
- This ensured robust evaluation of model performance across different data splits.

7.6. Model Implementation:

Four machine learning models were implemented for this project. The first was a Logistic Regression model, built within a pipeline that included standard scaling and class balancing to ensure fair treatment of the data.

The second model was a Random Forest Classifier, which combined 200 decision trees, each limited to a maximum depth of 10. Class weighting was applied to account for any imbalance in the dataset. Third was a Gradient Boosting Classifier, an ensemble of 200 estimators trained with

a conservative learning rate of 0.05 and a maximum tree depth of 5, allowing it to learn gradually without overfitting.

Finally, XGBoost which is like the Gradient Boosting model, it ran for 200 iterations at a learning rate of 0.05, but leverages histogram binning internally for faster and more efficient training.

7.7. Model Training:

- Models were trained using the `cross_validate()` function from scikit-learn.
- Stratified K-Fold ensured consistent class distribution during training.
- Training was performed separately for each model across 10 folds.

7.8. Model Evaluation:

- Performance was evaluated using multiple metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - ROC-AUC
- Confusion matrices were generated using cross-validated predictions.
- Sensitivity (Recall) and Specificity were also computed from confusion matrix values.

7.9. Clustering Implementation:

K-Means clustering was applied to a selection of climate and epidemiological features — temperature, precipitation, leaf area index, geographic coordinates, the rolling mean of cases, death rate, and the interaction term. Before clustering, all features were standardized to ensure no single variable dominated simply due to its scale.

The optimal number of clusters was determined by examining two things: the elbow method, which looks at how inertia drops as more clusters are added, and the silhouette score, which measures how well-separated the clusters actually are. DBSCAN was also explored as a complementary approach, particularly useful for identifying dense outbreak regions and flagging unusual data points as outliers.

Once the clusters were formed, they were analysed from a few different angles, looking at the outbreak rate within each cluster, comparing the average feature values across groups, and mapping them geographically using latitude and longitude to see whether the clusters had any meaningful spatial pattern.

Beyond clustering, heatmaps were used to visualise how disease spread varied across districts month by month, while time-series plots helped reveal whether outbreaks followed any kind of seasonal rhythm over time.

7.10 Early Warning System Implementation:

A trained XGBoost-style model was used to predict outbreak probability. The latest record for each district was extracted using sorting and grouping. Prediction probabilities were computed using `predict_proba()`. Districts with probability greater than 0.7 were classified as high-risk early warning zones.

7.10.1. Tools and libraries used for implementation:

- **Programming Language** : Python 3.x – Primary language used for data preprocessing, feature engineering, model building, clustering, and visualization.
- **Data Handling and Processing** : Pandas – Used for loading datasets, data cleaning, preprocessing, aggregation, and feature engineering. NumPy – Used for numerical computations and efficient array operations.
- **Machine Learning and Statistical Modeling** : Scikit-learn – Used for most machine learning tasks including:
 - Train-test splitting (`train_test_split`)
 - Preprocessing (`StandardScaler`, `LabelEncoder`)

- Classification models (Logistic Regression, Random Forest, Gradient Boosting, Histogram-based Gradient Boosting)
- Evaluation metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC)
- Cross-validation (Stratified K-Fold)
- Clustering algorithms (K-Means)
- Silhouette Score and Davies-Bouldin Index

XGBoost (conceptually equivalent via HistGradientBoostingClassifier) – Used for gradient boosting-based classification and outbreak probability prediction.

- **Data Visualization** : Matplotlib – Used for plotting graphs such as feature importance, time-series trends, cluster visualization, and model comparison charts. Seaborn – Used for heatmaps, statistical plots, and confusion matrix visualization. Matplotlib CM (colormap utilities) – Used for cluster color mapping and silhouette plots.
- **Model Evaluation and Validation** : Scikit-learn metrics module – Used for:
 - Confusion Matrix
 - ROC-AUC Score
 - Classification Report
 - Silhouette Score
 - Cross-validation evaluation metrics
- **Development Environment** : Jupyter Notebook / Google collab – Used for writing and executing Python code interactively. IPython Kernel – Used for real-time execution and debugging.

7.11. Code snippets or pseudocode

7.11.1. Data Preprocessing

```
df = load_dataset()

df['Cases'] = convert_to_numeric(df['Cases'])

df = sort_by(['district', 'year', 'month', 'day'])
```

```
df['Deaths'] = fill_missing(df['Deaths'], value=0)

df['Temp'] = convert_kelvin_to_celsius(df['Temp'])\
```

7.11.2. Feature Engineering

```
df = load_dataset()

df['Cases'] = convert_to_numeric(df['Cases'])

df = sort_by(['district', 'year', 'month', 'day'])

df['Deaths'] = fill_missing(df['Deaths'], value=0)

df['Temp'] = convert_kelvin_to_celsius(df['Temp'])
```

7.11.3. Target Variable Creation

```
if Cases > 10:

    Outbreak = 1

else:

    Outbreak = 0
```

7.11.4. Risk Score Calculation

```
Risk_Score = df_norm["Risk_Score"] = (w_cases * df_norm["Cases"] + w_preci *
df_norm["preci"] + w_temp * df_norm["Temp_C"])
```

7.11.5. Model Training

```
model = RandomForestClassifier(n_estimators=200, max_depth=10)

model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

7.11.6. K-Means Clustering

```
kmeans = KMeans(n_clusters=3)

clusters = kmeans.fit_predict(X_scaled)
```

7.11.7. Evaluation

```
accuracy = accuracy_score(y_test, predictions)
```

```
roc_auc = roc_auc_score(y_test, probabilities)
```

Chapter 8

Result

8. 1. Presentation of the results obtained from the implementation

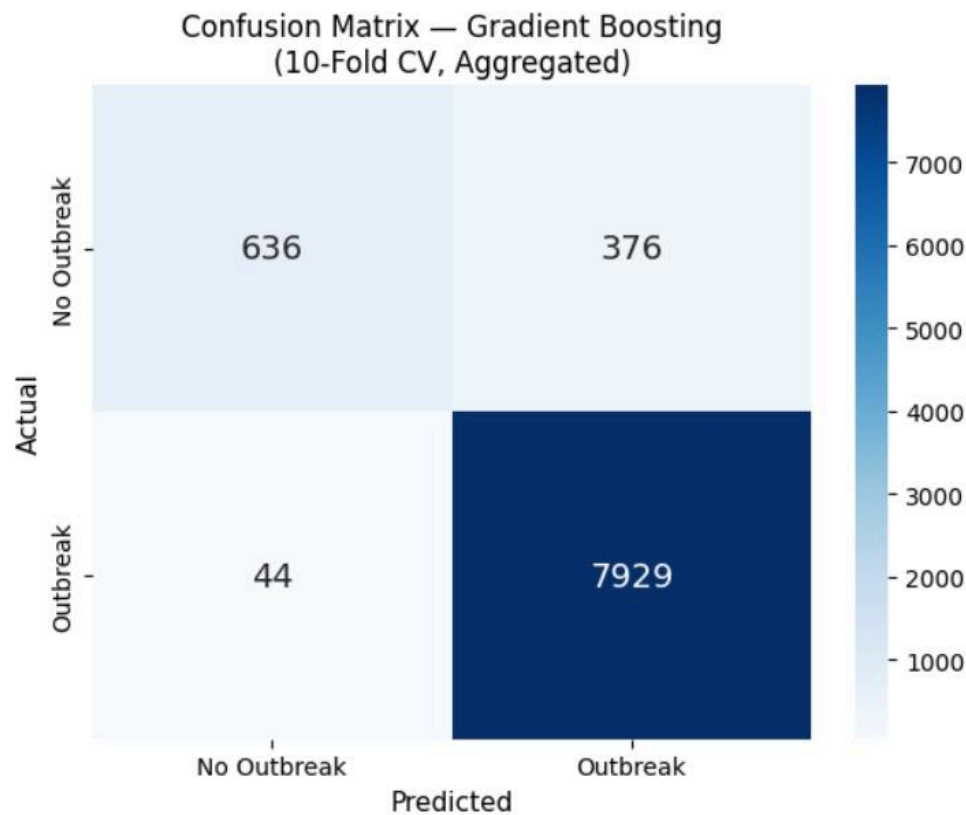


Figure 9.1: Table and Confusion matrix of the results obtained from the implementation

Figure 8.1 shows the confusion matrix representing the comparison between the actual and predicted values for the outbreak classification model (Gradient Boosting with 10-fold cross-validation).

In this case, each row represents the actual class, and each column represents the predicted class. The two classes considered are “No Outbreak” (0) and “Outbreak” (1).

From the confusion matrix:

0 0 (True Negatives): There were 636 instances where the actual label was “No Outbreak” and the model also predicted “No Outbreak”. This indicates that the model correctly identified non-outbreak cases.

1 0 (False Negatives): There were 44 instances where the actual label was “Outbreak”, but the model incorrectly predicted “No Outbreak”. These are critical errors, as actual outbreaks were missed by the model.

0 1 (False Positives): There were 376 instances where the actual label was “No Outbreak”, but the model incorrectly predicted “Outbreak”. These represent false alarms generated by the model.

1 1 (True Positives): There were 7,929 instances where both the actual and predicted labels were “Outbreak”. This indicates that the model successfully identified a large number of outbreak cases.

The confusion matrix provides valuable insights into the model’s performance. The high number of correctly classified outbreak cases (true positives) shows that the model is highly effective in detecting outbreaks. The relatively low number of false negatives (44) indicates that the model rarely misses actual outbreak events, which is crucial in epidemic prediction systems.

Overall, the strong diagonal values (636 and 7,929) compared to the off-diagonal values indicate that the model performs well in classification. The model achieves high recall (sensitivity), meaning it is very effective in identifying outbreak cases, though there is some trade-off with specificity due to false positives.

8.2.Evaluation Metrics Used to Assess Model Performance

To evaluate the performance of the machine learning models developed for predicting epidemic outbreak risk, multiple classification metrics were used. These metrics provide a comprehensive understanding of how well the models distinguish between outbreak and non-outbreak cases, especially under class imbalance conditions.

- **Accuracy:** Accuracy measures the proportion of correctly classified instances out of the total number of instances. It is defined as the ratio of correct predictions (both outbreak and non-outbreak) to the total predictions made by the model. While useful as a general indicator, accuracy alone is not sufficient in imbalanced datasets.
- **Precision :** Precision measures the proportion of correctly predicted outbreak cases out of all instances predicted as outbreaks. It is calculated as : $\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$
- **Recall (Sensitivity) :** Recall measures the proportion of actual outbreak cases that are correctly identified by the model. It is defined as: $\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$
- **F1 Score :** The F1 score is the harmonic mean of precision and recall. It provides a balanced measure between the two, especially when dealing with imbalanced datasets. $\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- **ROC Curve and ROC-AUC Score :** The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate at different classification thresholds. The Area Under the Curve (ROC-AUC) provides a single scalar value representing the model's ability to distinguish between outbreak and non-outbreak classes. A higher AUC indicates better classification performance.
- **Confusion Matrix :** A confusion matrix provides a detailed breakdown of model predictions, including: True Positives, True Negatives, False Positives, False Negatives. It is used to analyze the types of classification errors made by the model.
- **Cross-Validation Metrics :** To ensure robust performance evaluation, Stratified K-Fold Cross Validation ($K = 10$) was used. This helps in assessing model stability across different subsets of the dataset and reduces bias due to a single train-test split. For each fold, the following metrics were computed: Accuracy, Precision, Recall, F1-score,

ROC-AUC. The final performance was reported as the mean and standard deviation across all folds.

- **Specificity** : Specificity measures the proportion of correctly identified non-outbreak cases. It is defined as: $\text{Specificity} = \text{True Negatives} / (\text{True Negatives} + \text{False Positives})$. It helps in evaluating how well the model avoids false alarms.

8.3. Comparison of results with previous approaches

8.3.1. Comparison of Models

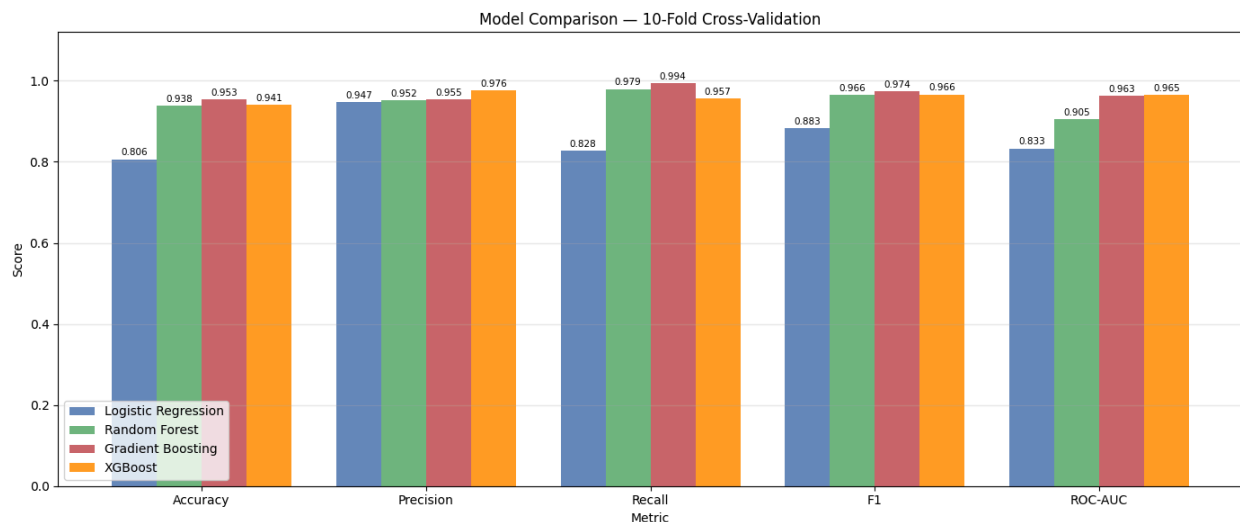


Figure 8.3.1: Model comparison

The Figure 8.3.1 shows us the model comparison between all the four models we trained and tested our data on

Logistic Regression did the worst out of the four. It got an accuracy of 0.806, so about 1 in 5 predictions were just wrong. For something like predicting disease outbreaks, that's not good enough.

Gradient Boosting (red) is the top performer overall. It leads in Recall (0.994), meaning it almost never misses an actual outbreak and has the highest F1 score (0.974), which reflects the best balance between catching outbreaks and avoiding false alarms. For a public health system, this matters most.

XGBoost is very very close behind and is actually a tiny bit better in Precision, meaning when it says there's an outbreak, it's usually right. It is nearly identical to Gradient Boosting and actually edges it out in Precision (0.976) and ROC-AUC (0.965), meaning it's slightly better at ensuring its positive predictions are correct and at distinguishing outbreak vs. non-outbreak cases overall.

Random Forest is also solid and not too far behind the other two. It performs pretty well but not as good as the Gradient and XGBoost even though it has a high accuracy of 93%. It just fell behind in the areas of ROC AUC Score.

So basically, Logistic Regression is out. The other three are all good good options, but Gradient Boosting is our top pick for this kind of problem.

From the results obtained through 10-fold cross-validation, the performance of different machine learning models was compared using key evaluation metrics such as F1-score and ROC-AUC.

Table 8.1: Model Comparison

Model	F1 Score	ROC AUC Score
Logistic Regression	0.8333	0.8330
Random Forest	0.9656	0.9053
Gradient Boosting	0.9742	0.9628
XGBoost	0.9662	0.9653

F1 Score:

The Gradient Boosting and XGBoost (Histogram-based Gradient Boosting) models achieved the highest F1-scores among all models, indicating a strong balance between precision and recall in detecting outbreak cases.

The Random Forest model also demonstrated competitive performance, with F1-scores close to the boosting models, making it a reliable alternative.

The Logistic Regression model showed comparatively lower F1-scores, indicating limitations in capturing complex non-linear relationships present in the dataset.

Overall, ensemble-based models (Gradient Boosting, Random Forest, and XGBoost) outperformed Logistic Regression in terms of F1-score, highlighting their effectiveness for epidemic outbreak prediction.

ROC-AUC Score:

In terms of ROC-AUC, the XGBoost (Histogram-based Gradient Boosting) and Gradient Boosting models achieved the highest scores, indicating strong capability in distinguishing between outbreak and non-outbreak cases.

- The Random Forest model also showed stable and high ROC-AUC values, demonstrating good classification performance.
- The Logistic Regression model had relatively lower ROC-AUC scores compared to ensemble models, reflecting its limited ability to handle complex feature interactions.
- Overall, boosting-based models (Gradient Boosting and XGBoost) demonstrated superior performance across both F1-score and ROC-AUC metrics. These models were more effective in identifying outbreak patterns due to their ability to capture complex relationships in the data.
- The Random Forest model provided strong and consistent performance, making it a robust alternative.

- The Logistic Regression model, while useful as a baseline, underperformed compared to ensemble methods.

Thus, ensemble learning techniques proved to be the most suitable approach for outbreak prediction in this study.

Chapter 9

Conclusion

This project focused on developing a data-driven framework for predicting epidemic outbreaks and identifying high-risk regions across India using epidemiological and climatic data. By integrating the EpiClim dataset with supplementary environmental features, a comprehensive analytical pipeline was designed and implemented.

Multiple machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and Histogram-based Gradient Boosting (XGBoost-like), were implemented and evaluated using stratified 10-fold cross-validation. Among these, ensemble-based models, particularly Gradient Boosting and XGBoost, showed superior performance in terms of F1-score, Accuracy and ROC-AUC, indicating their effectiveness in handling complex, non-linear relationships in outbreak data.

The confusion matrix analysis revealed that the models achieved high recall, meaning they were highly effective in detecting actual outbreak cases. This is especially critical in public health applications, where missing an outbreak can have serious consequences. Although some false positives were observed, they are generally more acceptable than false negatives in an early warning system.

In addition to classification, K-Means clustering was used to identify distinct epidemic risk patterns across districts. Visualization techniques further helped in understanding spatial and temporal trends in disease spread.

Overall, this project highlights the effectiveness of combining machine learning, statistical analysis, and domain knowledge to build a robust epidemic prediction system. The proposed framework can assist policymakers and healthcare authorities in proactive planning, resource allocation, and timely intervention, ultimately helping to reduce the impact of disease outbreaks.

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